

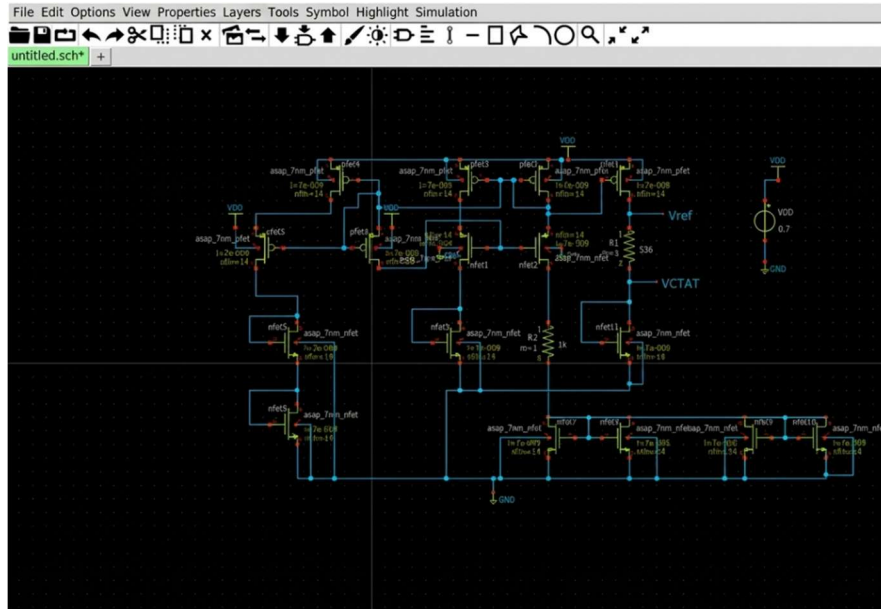
Module 3 Assignment

VSD 7nm FinFET Workshop

For my username, **mansi**, the ASCII sum = $109 + 97 + 110 + 115 + 105 = 536$

Therefore, Resistor Value : 536

1) The Xschem schematic

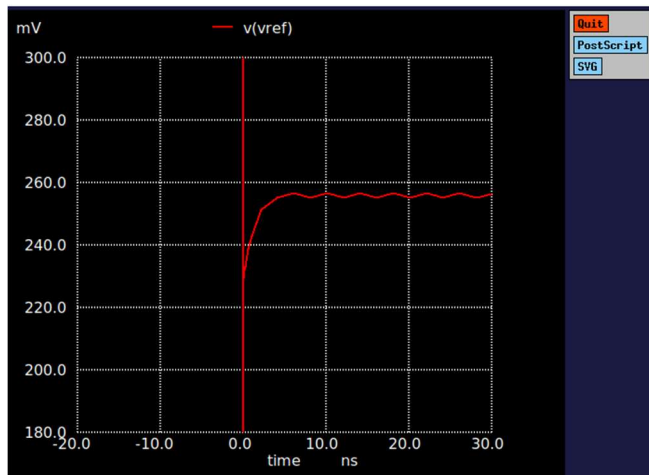


2) The Unique Resistor Value

```
*bandgap.spice
~/Desktop/Bandgap-Reference-Circuit-with-SCMB-with-ASAP-7nm-PDK-
Save

1 ** sch_path: /home/madhuri/asap_7nm_Xschem/bandgap_ref.sch
2 **.subckt bandgap_ref
3 Xnfet1 net3 net1 net7 net10 asap_7nm_nfet l=7e-009 nfin=14
4 Xnfet2 net2 net1 net9 net11 asap_7nm_nfet l=7e-009 nfin=14
5 Xpfet1 net3 net2 VDD net12 asap_7nm_pfet l=7e-009 nfin=14
6 Xpfet2 net2 net2 VDD net13 asap_7nm_pfet l=7e-009 nfin=14
7 Xpfet3 Vref net2 VDD net14 asap_7nm_pfet l=7e-009 nfin=14
8 Xpfet4 net4 net2 VDD net15 asap_7nm_pfet l=7e-009 nfin=14
9 Xpfet5 net5 net2 net4 net16 asap_7nm_pfet l=7e-009 nfin=14
10 Xpfet6 net1 net2 net2 net17 asap_7nm_pfet l=7e-009 nfin=14
11 Xnfet3 net5 net5 net6 net18 asap_7nm_nfet l=7e-009 nfin=14
12 Xnfet4 net6 net6 GND net19 asap_7nm_nfet l=7e-009 nfin=14
13 Xnfet5 net7 net7 GND net20 asap_7nm_nfet l=7e-009 nfin=14
14 Xnfet6 net8 net8 GND asap_7nm_nfet l=7e-009 nfin=14
15 Xnfet7 net8 net8 GND GND asap_7nm_nfet l=7e-009 nfin=14
16 Xnfet8 net8 net8 GND GND asap_7nm_nfet l=7e-009 nfin=14
17 Xnfet9 net8 net8 GND GND asap_7nm_nfet l=7e-009 nfin=14
18 R1 net9 net8 33k ac=1k m=1
19 R2 Vref Vctat 536 ac=1k m=1
20 Xnfet10 Vctat Vctat GND net21 asap_7nm_nfet l=7e-009 nfin=14
21 V2 vdd GND 1
22 **** begin user architecture code
23
24 ** VDD vdd 0 PULSE(0 1 0n 1p 50n 100n)
25 ** .temp 27
26 ** .tran 10n 100n
27
28 .dc temp -45 125 5
29 .control
30 run
31 plot v(Vref)
32 plot v(Vctat)
33 plot v(Vref)-v(Vctat)
34 plot v(Vref) v(Vctat)
35 plot v(Vctat) v(Vref)-v(Vctat)
36 let temp_coeff = deriv(v(Vref))/1.24
37 plot temp_coeff
38 plot abs(Iv2#branch)
39 measure dc Vref FIND v(vref) WHEN TEMP=125
40 print Vref
41
42 .endc
43
44
```

3) Transient Waveform Showing Startup Time



4) Characterization Table

S. No	VDD (V)	Temp	Vref	Line Reg (mV / V)	Startup Time (ns)
1	0.8	27	214 mV	266.2	5.49
2	0.9	27	235mV	260	6.12
3	1.0	27	257mV	256	6.40
4	1.0	-40	273mV	271	63
5	1.0	125	234mV	236	0.516

```

DC transfer characteristic Fri May 8 22:22:38 2026
-----
Index  temp-sweep  vref
-----
0      -4.00000e+01  2.288555e-01
1      -3.90000e+01  2.286323e-01
2      -3.80000e+01  2.284078e-01
3      -3.70000e+01  2.281830e-01
4      -3.60000e+01  2.279582e-01
5      -3.50000e+01  2.277332e-01
6      -3.40000e+01  2.275081e-01
7      -3.30000e+01  2.272828e-01
8      -3.20000e+01  2.270575e-01
9      -3.10000e+01  2.268320e-01
10     -3.00000e+01  2.266064e-01
11     -2.90000e+01  2.263808e-01
12     -2.80000e+01  2.261550e-01
13     -2.70000e+01  2.259292e-01
14     -2.60000e+01  2.257033e-01
15     -2.50000e+01  2.254773e-01
16     -2.40000e+01  2.252513e-01
17     -2.30000e+01  2.250252e-01
18     -2.20000e+01  2.247990e-01
19     -2.10000e+01  2.245728e-01
20     -2.00000e+01  2.243466e-01
21     -1.90000e+01  2.241204e-01
22     -1.80000e+01  2.238941e-01
23     -1.70000e+01  2.236678e-01
24     -1.60000e+01  2.234415e-01
25     -1.50000e+01  2.232151e-01
26     -1.40000e+01  2.229888e-01

```

VDD = 0.8, Temp = 27

66	2.600000e+01	2.140082e-01
67	2.700000e+01	2.137873e-01
68	2.800000e+01	2.135667e-01

VDD = 0.9, Temp = 27

66	2.600000e+01	2.354158e-01
67	2.700000e+01	2.351916e-01
68	2.800000e+01	2.349678e-01

VDD = 1, Temp = 27

66	2.600000e+01	2.573412e-01
67	2.700000e+01	2.571127e-01
68	2.800000e+01	2.568846e-01

VDD = 1, Temp = -40

0	-4.000000e+01	2.730239e-01
1	-3.900000e+01	2.727823e-01
2	-3.800000e+01	2.725395e-01

VDD = 1, Temp = 125

163	1.230000e+02	2.380184e-01
164	1.240000e+02	2.378623e-01
165	1.250000e+02	2.377075e-01

5) DC Analysis Plots

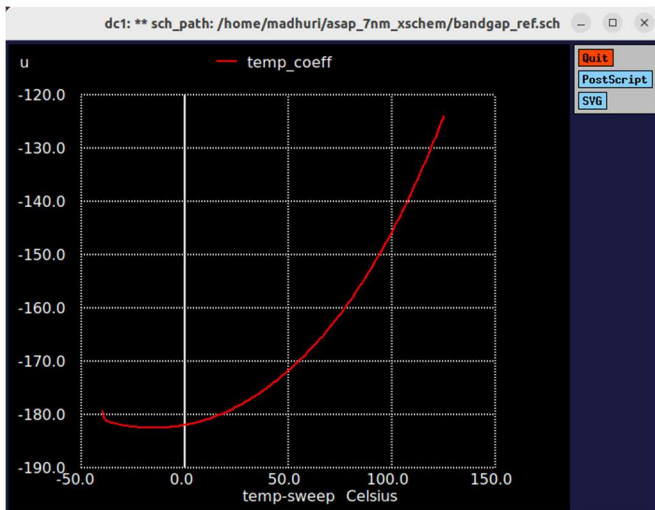


Fig 1. Temperature Coefficient

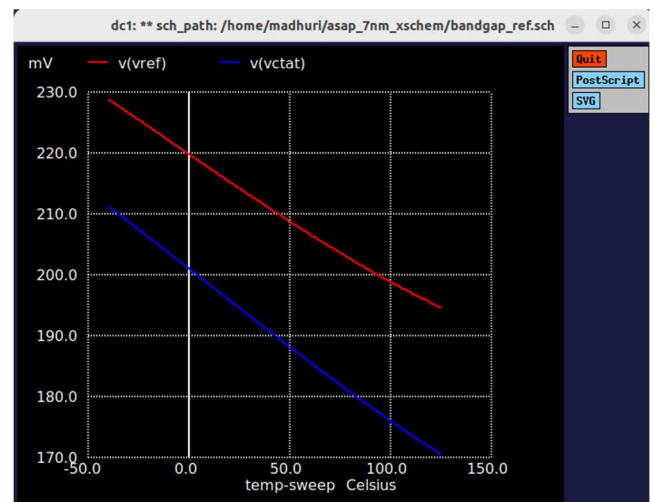


Fig 2. VREF and CTAT

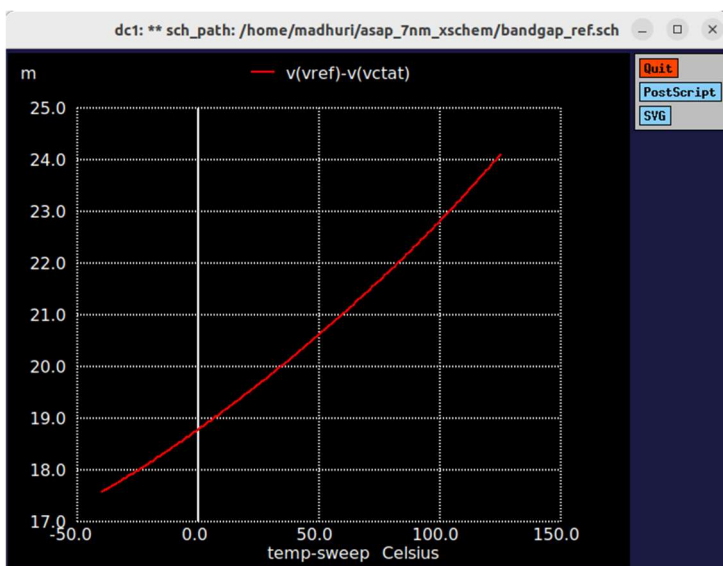


Fig 3. PTAT